NANNY-APP · PRODUCT & ENGINEERING

Implementation Plan

Phased delivery, CI/CD, deployment, testing and maintenance

Version 1.0 10 October 2026 Status: implementation baseline + launch requirements

1. Current baseline

The repository already contains the PHP web portal, Android app, shared business services, MariaDB migrations, private storage abstraction, automated synthetic checks, an APK download endpoint, an admin action centre and AWS pilot configuration. Local PHP/JavaScript/PWA/cache/download checks, disposable MariaDB migration/booking regressions, HTTP security checks, Android assembly, seven JVM tests and lint passed on 10 October 2026.

> **Open evidence:** the AWS stack is not created; the production container build was blocked locally by Docker clock drift; live mail, DNS/TLS, backups, signed APK and physical-device acceptance remain unverified.

2. Delivery phases and indicative timeline

Phase 0 — Decisions and accounts · 1–2 days

Confirm domain, owner/operator, support details, email sender, retention policy, recovery objectives and AWS budget alarms. Use Stockholm `eu-north-1`.

Phase 1 — Production hardening · 2–4 days

Resolve Docker clock, build image, review environment defaults, validate CloudFormation, remove demo data, approve legal/safety copy and complete remaining release review.

Phase 2 — AWS staging · 2–3 days

Create EC2/EIP/security group, DNS, TLS, database volumes, secrets and workers. Run migrations and synthetic smoke tests.

Phase 3 — Release validation · 3–5 days

Restore drill, live email, private media, browser/device/accessibility, concurrency and rollback tests. Fix defects and rerun gates.

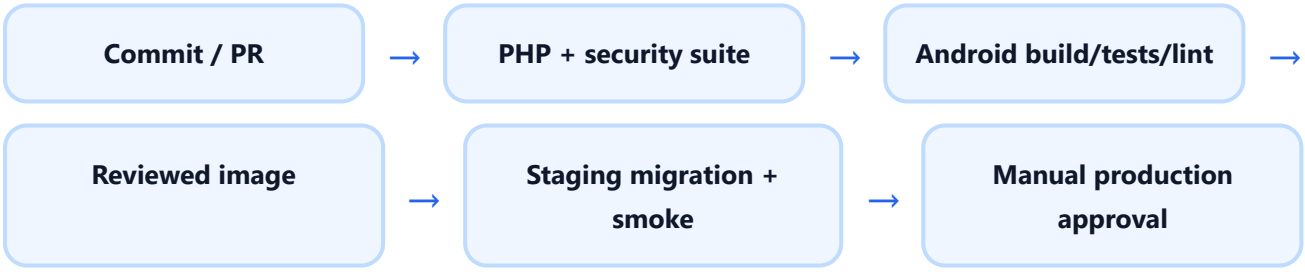
Phase 4 — Android release · 1–2 days

Build with real HTTPS API URL, sign with protected key, verify SHA-256/install/update and place APK in private release folder.

Phase 5 — Controlled pilot · 2 weeks

Invite a small cohort, monitor queues/errors/feedback daily and review success metrics weekly.

3. CI/CD pipeline



- GitHub Actions runs isolated MariaDB/HTTP/mail/security tests and Android verification. Resolve the recorded GitHub account billing lock before relying on hosted CI.
- Build immutable application image; keep credentials, uploads, APK signing key and runtime data outside it.
- Back up and restore-test before migrations. Run only the shared CLI migration runner.
- Deploy staging, test, then switch/restart deliberately. Keep the previous image and additive schema for rollback.
- Do not automatically deploy on every push during the pilot.

4. Test strategy

Layer	Coverage	Release gate

Static	PHP/JS syntax, Android lint, Compose compilation, workflow validation.	No errors
Unit	Android model/session cache boundaries and shared pure logic.	All pass
Integration	Migrations, bookings/ledger, authorization, media, mail/outboxes, rate limits.	Disposable DB suite passes
End-to-end	Parent, nanny and admin smoke journeys over staging HTTPS.	All critical journeys pass
Device/accessibility	Android small/large phones; Safari PWA; desktop/mobile web; keyboard/screen reader/zoom.	No critical/serious blockers
Recovery	Restore DB and uploads into isolated environment; verify integrity.	Measured RPO/RTO accepted

5. AWS deployment runbook

1. In Stockholm, upload `deploy/aws/ec2-stack.yml` to CloudFormation; review costs before creating.
2. Point the domain A record to the Elastic IP and restrict SSH to the operator's `/32`.
3. Install Docker on Ubuntu, copy reviewed source to `/srv/nannyapp` and create protected `deploy/aws/.env`.
4. Build; start MariaDB; run migrations; start app, Caddy and workers.
5. Verify TLS, login, booking, messages, uploads, recovery mail, queues and admin workflows.
6. Back up, restore-test, enable external checks and alerts, then admit synthetic users.
7. Build/sign the APK with `https://DOMAIN/api/` and publish it as `deploy/aws/releases/NannyApp-latest.apk`.

6. Operations and maintenance

Daily

HTTPS, 5xx, disk, DB, worker queue age, mail failures, safety tickets and disputes.

Weekly

Backups, dependency/security notices, support trends, KPI review and test suite.

Monthly

Restore sample, access review, patching, cost/credit review and capacity trend.

Per release

Backup, migration rehearsal, smoke test, APK checksum/signature, rollback image and release record.

7. Risks and ownership

Risk	Mitigation	Owner
Credits end / cost overrun	Budgets/alerts, weekly credit review, export before 18 Mar 2027.	Account owner
Single-host outage	Off-host backups, restore drill, documented replacement procedure.	Operator
Private-data exposure	Authorization tests, private paths, no public bucket, log hygiene.	Engineering + operator
Manual-payment misunderstanding	Consistent copy, reconciliation SOP, no escrow claims.	Product + admin
Signing-key loss	Encrypted offline backup, named owner and recovery record.	Release owner

8. Definition of done

Production is ready only when code checks pass, CloudFormation and image are validated, HTTPS/domain/email/workers function, migrations and restore are proven, private uploads are authorized and backed up, parent/nanny/admin staging journeys pass, signed APK installs on real devices, policies/support contacts are approved, monitoring is active and rollback has been rehearsed.

Document suite